

ODY SHBAYEH

Software Engineer

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SUMMARY

Software Engineer with hands-on experience shipping production systems at RIG Holding, spanning backend services (Spring Boot, .NET), real-time communication (WebRTC, SIP, RTSP), and full-stack web/mobile applications. Computer Engineering graduate (Birzeit University) with additional backend and DevOps training on Microsoft Azure. Comfortable owning a feature end-to-end, from API design through deployment.

EXPERIENCE

Software Engineer — RIG Holding, Ramallah *Oct 2025 – Present*

Deliver enterprise-grade services integrated with the company's core monolithic platform.

- Built a CRM landing page builder that lets the marketing team launch and manage campaign pages without engineering support, synced in real time with the core platform for campaigns, customer interactions, and business processes.
- Developed a VIP Lounge platform providing secure live RTSP camera access via automated subscriptions, with real-time voice/video to service personnel using WebRTC, SIP, and WebSocket Secure (WSS) over an Asterisk backend.
- Built an internal Trade-In management application handling vehicle evaluation workflows, inventory tracking, pricing decisions, approvals, and archival for the operations team.
- Designed and built a wrapper API around Meta's GraphQL Messaging API to centralize CRM messaging across the company, now being integrated as the messaging layer for the RIG Automotive mobile app.
- Currently developing the RIG Automotive mobile app, an aftersales platform for service requests, appointment scheduling, and parts requests.

DevOps Trainee — ASAL Technologies, Ramallah *Feb 2025 – Jun 2025*

- Trained in Microsoft Azure cloud infrastructure, deployment automation, and release workflows.
- Built CI/CD pipelines and managed source control using Azure DevOps, improving repeatability of deployments.

Backend Engineering Trainee — ASAL Technologies, Ramallah *Sep 2024 – Feb 2025*

- Developed and maintained RESTful APIs in Java (Spring Boot), implementing full CRUD functionality against MongoDB.
- Collaborated with cross-functional teams to design and maintain scalable backend services.

ACADEMIC & PERSONAL PROJECTS

Intelligent Traffic Management & Red-Light Violation System — Capstone, Birzeit University *2024 – 2025*

- Processed 30-minute, 4K/30fps drone footage using YOLOv8 + SORT tracking to detect vehicles and traffic-light states in real time, achieving 86% red-light-violation detection accuracy.
- Calculated congestion metrics from polygonal ROI vehicle counts and recommended adaptive green-light durations, simulating a 25% reduction in rush-hour queue length.
- Logged frame-level JSON summaries and evidence images to MongoDB, exposed via a FastAPI backend to a React dashboard.

Repo: github.com/odyshbayeh/TrafficManagementRedLightViolationDetection

Trainee's Portal — Backend Developer — Spring Boot, MySQL, Postman

- Co-built a 2-person team project enabling ASAL trainees to browse and enroll in training courses.
- Developed RESTful APIs in Java/Spring Boot with MySQL, including search and filtering.
- Implemented form validation, context-based state management, and end-to-end authentication/authorization.

Additional repos: [Automotive Database System — Java/JavaFX + MySQL](#) · [CPU Simulation \(Real-Time Systems\)](#) · [UVM Hardware Timer Module \(SystemVerilog\)](#) · [Bungee Jumping Simulation \(Embedded C\)](#) · [Spy/Espionage Simulation \(Embedded C\)](#)

TECHNICAL SKILLS

Languages: Java, C#, Python, JavaScript/TypeScript, C, Dart

Backend & APIs: Spring Boot, .NET, RESTful & GraphQL APIs, Microservices Architecture, Authentication & Authorization

Frontend & Mobile: React, Next.js, Responsive UI Development, Flutter, Cross-Platform Development

Cloud & DevOps: Microsoft Azure, Azure DevOps, CI/CD Pipelines, Docker, Git/GitHub/GitLab

Real-Time Communication: WebRTC, SIP, WebSockets, Asterisk, RTSP

Databases: SQL, MongoDB, MySQL, Oracle

Also familiar with: Verilog, MIPS/Intel Assembly, UVM (coursework)

EDUCATION

B.Sc. Computer Engineering, Birzeit University

July 2025

Relevant coursework: Operating Systems, Database Systems, Data Structures, OOP, Embedded Systems, Software Engineering

Certifications: English C1 Advanced — EF SET Certificate ([verify](#))